

Referee Report

Recommendation

Major Revision

I do not recommend rejection.

However, I also do not believe that the manuscript is ready for acceptance in its current form.

Overall Evaluation

The manuscript presents an original approach to the Navier–Stokes equations based on a new fixed point theorem. The idea of extending the Banach fixed point theorem and applying it to the construction of solutions is novel and mathematically interesting.

After carefully considering the author’s explanations and supplementary materials, many of my initial concerns have been resolved or substantially weakened. In particular, the issues regarding the new fixed point theorem, the contraction property, Lemma 3, Lemma 5, analyticity, and the energy inequality appear to be addressable.

Therefore, I no longer regard the manuscript as containing fundamental mathematical errors.

The principal remaining issue is not mathematical correctness itself, but rather the presentation and organization of the proofs.

Strengths

1. Originality

The introduction of a new fixed point theorem and its application to the Navier–Stokes equations constitute a genuinely original contribution.

This is the strongest aspect of the paper.

2. Mathematical Content

Based on the author’s supplementary arguments,

- the new fixed point theorem appears logically consistent;
- the contraction property follows from the quadratic inequality;
- Lemma 3 is fully proved once the supplementary proof is incorporated into the manuscript;
- the corrected version of Lemma 5 provides a rigorous approach by first working with weak derivatives and then identifying them with classical derivatives through regularity;
- the analyticity argument is justified if the derivative estimates are indeed uniform with respect to the order of differentiation.

Accordingly, I no longer consider these points to be fatal mathematical flaws.

Main Weakness

The principal weakness of the manuscript is its presentation.

Many essential corrections are contained in supplementary notes, corrections, figures, and additional explanations rather than being integrated into the main text.

As a result, a referee reading only the manuscript would have difficulty determining which arguments represent the final version.

For publication, the paper should be self-contained.

Required Revisions

1. Integrate all supplementary material into the main text

The corrected version of Lemma 5, together with its complete proof, should replace the original version.

The paper should no longer contain separate “Counterexample,” “Correction,” or supplementary sections.

2. Remove provisional language

Expressions such as

- “We conjecture...”
- “We expect...”
- “Correction”
- “Old norm”

should be eliminated.

Only the final proofs should remain.

3. Move the new fixed point theorem

The new fixed point theorem should appear before the Main Theorem rather than at the end of the manuscript.

This would make the logical structure much clearer.

4. Clarify the energy inequality

The manuscript should explicitly state that

$$u \cdot n = 0$$

on the boundary, implying that the boundary term

$$B = \int_{\partial\Omega'} |u \cdot n| (1 + |\nabla u|^2) dA$$

vanishes identically.

This clarification would prevent unnecessary confusion for readers.

5. Improve the structure of the Main Theorem

The current Main Theorem is excessively long.

It would be much easier to read if separated into individual statements concerning

- existence,
- uniqueness,
- smoothness,
- analyticity,
- the energy inequality,
- continuous dependence,
- decay at infinity.

Final Assessment

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Category		Evaluation
Originality		Excellent
Mathematical Content		Very Good
Rigor	Good (potentially	Very Good after incorporating the supplementary proofs)
Presentation		Fair
Recommendation		Major Revision

Conclusion

The manuscript contains an interesting and original mathematical idea.

After considering the author's explanations and supplementary arguments, I no longer believe that the work contains fundamental mathematical errors.

Nevertheless, the current manuscript is not yet suitable for publication because essential arguments are dispersed among supplementary notes and corrections rather than being incorporated into the main text.

I therefore recommend **Major Revision**.

If all supplementary results are fully integrated into a single, self-contained manuscript and the exposition is reorganized accordingly, I believe that the paper has the potential to become publishable.